

How multilingual setups affect accuracy in detecting the genre in news articles

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Abstract

We propose a multilingual approach to news genre categorization, specifically distinguishing between reporting, opinion and satire. An incremental training strategy is employed to evaluate the models performance as we scale from monolingual to multilingual setups to observe how cross-lingual knowledge affects accuracy. This establishes a robust analytical layer for media monitoring, preventing the misinterpretations of content in global news streams. The results of the paper show a nonlinear increase of the model performance during the iterative process. Because of the dataset size limitations and small linguistic diversity further work is required to verify the results in scale.

1. Introduction

Words used in news media play a crucial role in shaping public beliefs and opinions. Therefore, it is essential to distinguish between different genres of news. In this work, we focus on a multi-class classification task at the article level, where the goal is to determine if the news article is an opinion piece, objective news reporting or satirical text (Piskorski et al., 2023a).

This problem is particularly important because large language models (LLMs) are increasingly used in media analysis tasks and text mining, where their output directly affects how information is understood and processed. Accurate genre classification is crucial, because it affects many downstream tasks such as fact-checking, summarization, information retrieval and bias analysis, where errors can lead to misleading and even harmful results (Bender et al., 2021).

A multilingual approach is essential in this context, as news content is produced in a wide range of languages and cultural settings. These variations can significantly affect how well the model is able to generalize (Joshi et al., 2020). This is also very important for practical systems, as they need to work well despite the language in the input text.

We aim to explore how training a model on different language setups affects its performance across other languages. Our goal is to better understand how models capture and transfer knowledge in multilingual news classification settings.

1.1. Related Work

Current research in news genre classification has either focused on tuning monolingual models for high accuracy on a single language or using highly multilingual models in a zero-shot capacity to extend model capabilities on cross-language transferring without the drop of accuracy (Piskorski et al., 2023b). While it was shown that multilingual pre-training improves cross-lingual transfer (Conneau et al., 2020), the incremental addition of languages during the task specific fine

tuning is an unexplored aspect of generalization capabilities.

There is a significant knowledge gap regarding the optimal diversity of languages in the finetuning phase. It is not known whether a model’s performance improves linearly as more training languages are added or if there is a threshold after which “knowledge loss” occurs (Khelli et al., 2025). To the best of our knowledge our work is the first to measure the difference in performance by comparing monolingual, bilingual, trilingual and multilingual configurations on the SemEval-2023 Task 3 dataset.

2. Methodology

2.1. Data Construction and Enhancement Strategy

The dataset used in this work is from the SemEval 2023 competition (Task 3), which consists of news articles annotated for genre across nine languages: English, French, German, Georgian, Greek, Italian, Polish, Russian and Spanish. A key challenge in this dataset is the imbalance in the number of available training data and the low number of data available, where the most prominent language English had 433 training examples and other languages had all less than 160.

To combat this, we adopt and adapt data augmentation strategies inspired by previous SemEval-2023 Task 3 solutions. The first strategy is generating data using sentence-level sampling, where 30% of sentences from each article are randomly selected and recombined to form new training instances (Sadeghi et al., 2023). The second is to generate synthetic articles per genre using language-specific prompts. Specifically, we utilized the `gpt-4o-mini` model to generate 200 synthetic articles per language for English, French, German and Russian, uniformly distributed across the three target genres. The generation process executed language-specific prompts instructing the model to write professional news content of at least 350 words per article to reflect

local language and style differences.

To evaluate the effectiveness of different data augmentation strategies, we also compared three training setups: (1) real data only, (2) real data augmented using sentence-level sampling (30% of sentences per document) (3) real data augmented with synthetically generated articles and (4) real data with a combination of synthetic and augmented data.

2.2. Model and Embedding Representation

For the core feature extraction pipeline, we employ the pre-trained multilingual transformer model, XLM-RoBERTa (xlm-roberta-base), to generate dense semantic representations of the news articles. Each document is mapped into a fixed-size embedding that captures its meaning across languages.

Standard transformer models can only handle sequences up to $M = 512$ subword tokens. Since news articles are often longer than this, simply truncating the text would discard important information, such as writing style and structure, which are useful for detecting genres like *satire*. To mitigate this constraint, we implement a block-based sentence pooling embedding strategy.

The pipeline begins by segmenting the raw input article into individual sentences using the NLTK tokenization framework. These sentences are separated into blocks, ensuring that the total subword token count within any single block does not exceed $M - 2 = 510$ tokens. This constraint strictly preserves two slots for the mandatory special tokens: the sequence-initiating [CLS] token and the sequence-terminating [SEP] token.

Let $B = \{b_1, b_2, \dots, b_n\}$ represent the set of compiled text blocks for a given article. Each block b_i is fed independently into the XLM-RoBERTa encoder. Once all block representations are assembled, the final unified vector representation for the entire news article, denoted as $\mathbf{x} \in \mathbb{R}^{768}$, is computed by taking the mean pooling across the collective block dimensions:

$$\mathbf{x} = \frac{1}{n} \sum_{i=1}^n \mathbf{e}_{b_i} \quad (1)$$

2.3. Classifier Architecture

The generated document vector $\mathbf{x}$ serves as the direct input to a deep feed-forward neural network classifier designed to predict the final text genre. The classifier’s architecture is structured as follows:

An input linear transformation layer projects the initial embedding space into a dense hidden state space ($\mathbb{R}^{768} \rightarrow \mathbb{R}^{512}$). This is followed by a sequence of $L = 5$ dense hidden layers, where each hidden layer maintains a uniform width of 512 units. A stabilization and activation block is integrated within each hidden layer, which consists of a 1D Batch Normalization layer paired with an Exponential Linear Unit (ELU) activation function. The batch normalization ensures stable gradient flow and regularizes the weights, while the ELU activation avoids vanishing gradients during

backpropagation. Finally, an output linear layer maps the final hidden states into the target label space $\mathbb{R}^3$, representing our three categorical genres: *satire*, *opinion*, and *reporting*.

The network outputs raw logits, which are mapped to a distinct probability distribution $\hat{\mathbf{y}}$ via a Softmax activation function. Optimization is performed by minimizing the Negative Log-Likelihood loss ($NLLLoss$) driven by the Adam optimizer. To prevent mathematical instability when executing log-transformations on zero-bound probabilities, a minor smoothing parameter $\epsilon = 10^{-9}$ is factored into the loss computation:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=0}^2 y_{i,k} \log(\hat{y}_{i,k} + \epsilon) \quad (2)$$

where N is the batch size, $y_{i,k}$ represents the binary ground-truth indicator for article i and class k , and $\hat{y}_{i,k}$ represents the predicted probability assigned by the classifier network.

3. Evaluation Setup

All experimental configurations are evaluated using two core performance metrics: Overall Accuracy and the Macro-averaged F_1 -score. The Macro F_1 -score is used to treat all three target classes (*satire*, *opinion*, and *reporting*) with equal importance. This prevents the majority class from inflating the performance metrics and ensures an accurate evaluation across imbalanced categories. To track localized performance and linguistic variations, both metrics are computed globally for the combined groups and separately for each individual language.

Due to the lack of gold labels for the test set, all evaluations were made on the development data. To evaluate the model’s generalization capabilities and cross-lingual transfer, we divided the languages into "seen" and "unseen" groups:

1. **"Seen" languages:** English, French, German, and Russian. These languages are present during the fine-tuning phase and evaluate performance on known languages and how well the model learns genre-specific features.
2. **"Unseen" languages:** Italian and Polish. These are not used during training, so we can test how well the model works on new languages and whether it learns language-independent features.

Performance is documented and compared across both groups to analyze overall results as well as performance for each language.

4. Research questions

To provide a new insight into the fine-tuning phase of multilingual models, this paper observes changes in model performance addressing research questions:

RQ1: How does a model performance change with the incremental addition of languages in the

fine-tuning, task-specific (news genre classification) dataset. Contribution: We report model performance changes on models with different number of languages in the fine-tuning phase of News genre classification tested on a set of "unseen" language examples.

RQ2: Does a model performance on a "seen" language drop with the incremental addition of languages in the fine-tuning, task-specific (News genre classification) dataset? Contribution: We report model performance changes on models with different number of languages in the fine-tuning phase of News genre classification tested on a set of one "seen" language examples that was present in every News genre classification dataset.

RQ3: How do different data augmentation strategies (sentence-level sampling vs. synthetic data generation) influence model performance across languages? Contribution: We report model performance changes on "seen" and "unseen" languages on different data augmentation strategies.

5. Results

5.1. Data enhancement method

For these results we will not include figures, but they were obtained with the same method as the incremental language experiments. The main comparison was between 4 setups: no data enhancement, synthetic data, augmented data, and a combination of augmented and synthetic data. The general trend was a drop of performance when adding synthetic data, an increase in performance when switching to just augmented data and a similar score between a combination (synthetic + augmented) and a pure augmentation method. Let's look at one distribution of Macro F1 scores:

No data enhancement	0.4462
Synthetic data	0.4434
Augmented data	0.4450
Combination	0.4557

In this example the combination exceeds all other results but usually the augmented data is the best choice of enhancement especially if outlier results are ignored. Even in the case of augmented data being worse than no enhancement we still need a way to increase the number of articles in the dataset to enable the correct experimental setup for research questions 1 and 2. This is why we chose to work with the data augmentation method.

5.2. Iterative language addition results

The results as shown on figures 1-3 depict the increase in performance during the iterative addition of languages in the post-train dataset across 5 seeds as well as the average of all seeds on a certain setup. Figure 1 shows the iterative increase when testing on one seen language (English) and answers research question 2. Figure 2 shows the increase when testing on unseen languages (It, Po).

we provide the increase of performance while testing on the whole test of seen languages after every iteration (shown on Figure 3). This is included to show impact on cross lingual transfer between seen languages.

The performance of the model on all experimental setups increases from start to finish of the iterative process but does so in a nonlinear fashion. For example, Figure 3, seed 4 shows a change in both directions as languages are iteratively added. The average of all seeds on all figures also shows a nonlinear increase.

We must note that these results cover only a small set of the iterative process due to dataset limitations and a lack of curated linguistic examples; further research must be conducted to find definitive knowledge. As for now, we can say that the nonlinearity of the increase in performance is present in this problem, which is a proof of concept we wanted to achieve within this paper.

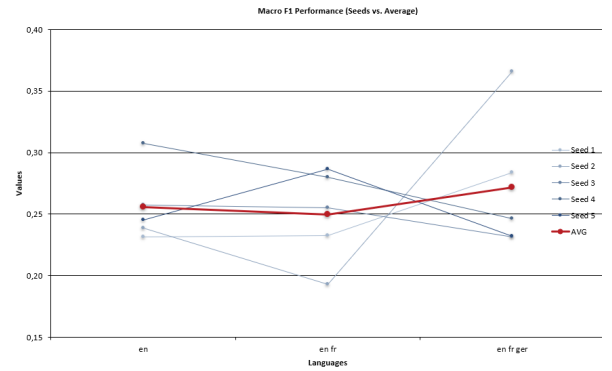


Figure 1: Testing on English

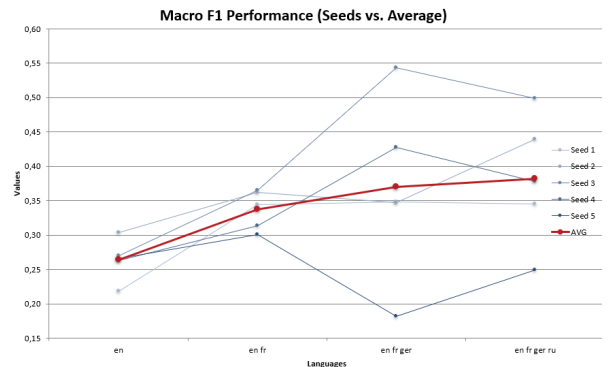


Figure 2: Testing on unseen languages (It, Po)

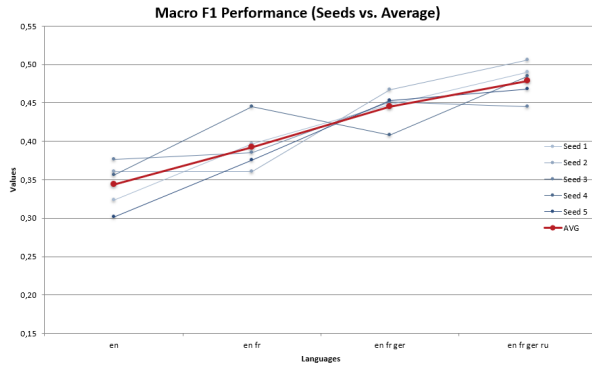


Figure 3: Testing on all seen languages (En, Fr, Ge, Ru)

6. Limitations

Due to the absence of gold labels for the official test set, all evaluations were performed on the development data. This may result in overly optimistic performance estimates.

Furthermore, the dataset is relatively limited in both size and linguistic diversity. It includes a restricted number of languages and does not adequately represent a wide range of linguistic families, which may limit the generalisability of the findings.

The hardware we used was also limited to one NVIDIA GeForce RTX 3050 Laptop GPU (4 GB).

Each experiment took between 10 and 20 minutes to complete using this setup.

7. Conclusion

The results of this paper show three key findings. The first is that the best method (among the ones that were tested) with the least introduction of error or even a small enhancement of performance is the data augmentation method.

The results also show a nonlinear improvement of performance across multiple experimental setups per seed showing the need to research this question in more detail on larger datasets, datasets with more linguistic diversity and larger end set of incremental languages. We leave these considerations of the problem for future work because of our limited data.

The average F1 score across multiple seeds shows an increase in performance but the increase drops towards the end also hinting at the nonlinearity of the improvement.

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